

Lesson 9 — Multiple Events at Once

Matches LEGO Teacher Portal lesson: "Rhythm Rescue".

Objective (I can...): I can set up more than one event so my model can respond in different ways. **CS concept:** Parallel/multiple event handlers · **Builds on:** Lesson 8 (sensor events).

AI Connection (2 min): A smart home watches many things at once — the door, the temperature, your voice. *First-principles question:* "How can one system pay attention to several events at the same time?"

Materials: LEGO CS & AI kit + device per pair, trigger objects/color cards. **Model reuse:** Events model from Lesson 6 — code only.

60-minute flow - Warm-up (8): "Many buttons" — coach calls out mixed triggers rapidly; pairs shout the matching response. - **Concept + demo (10):** Build two separate event stacks (a tap and a sensor) in one program; show both work independently. - **Build/Code (30):** Pairs program at least two independent event stacks (tap → spin; object near → sound) on the built model. *Swap at 15 min.* - **Challenge (7):** Add a third event; test triggering them in different orders. - **Share + pack-up (5):** Pairs demo each event firing on its own. Model stays built.

Pair roles: Each partner owns one event stack; swap at 15 min to review each other's. (*Trio: three stacks, one each.*)

Differentiation: *Support* — a worked twin event stack to copy the pattern from. *Extension / fast-finisher* — trigger two events nearly at once and observe. *Reteach* — build one stack, test it, then duplicate and change it for the second.

Exit check: Pair's program contains two+ event stacks that each work when triggered. **At-home (optional):** Draw a machine that responds to 3 different events. ~10 min.
